

I. Introduction

Write a brief overview of the analysis. State the primary objective, why this problem matters, and provide a quick summary of the dataset being used.

II. Data Preprocessing (Core Focus)

This section should detail every step taken to clean and prepare the data before any analysis begins. Do not skip steps; show your work.

- **Initial Data Inspection:** Check for missing values, duplicated data, and verify data types.

Example: Use `data.duplicated().sum()` to confirm there were 0 duplicate rows and confirmed correct data types (e.g., `int64` for scores).

- **Identify & Handle Outliers:** Use visualizations like boxplots to find extreme anomalies. Explain how you handled them.

Example: Boxplots revealed extreme outliers in hours worked (e.g., below 20 and up to 80). These were handled using the Interquartile Range (IQR) method to make the data suitable for modeling.

- **Feature Engineering:** Explain any transformations, encodings, or new variables you created.

III. Exploratory Data Analysis (Core Focus)

Use visualizations to uncover patterns, trends, and relationships in the clean data. This should be the longest section of your report.

- **Univariate Analysis (Distributions):** Look at variables one at a time using histograms or density plots.

Example: Histograms showed that "Hours Worked Per Week" had a roughly bell-shaped distribution centered around 40 hours with a slight right skew.

- **Bivariate Analysis (Relationships):** Compare variables against one another to find initial insights. Use scatter plots or grouped boxplots.

Example: A scatter plot with a best-fit line revealed that remote workers tend to have higher well-being scores and work fewer hours (20-40 hours) compared to in-office workers (up to 60 hours).

- **Correlation Analysis:** Use a correlation matrix/heatmap to quantify the linear relationships between all numerical variables.

Example: The heatmap showed a moderate negative correlation (-0.6) between Employment Type and Hours Worked, proving remote workers log fewer hours.

IV. Model Development (Secondary Focus)

Briefly explain the models chosen to predict your target variable. Since the focus is on data comprehension, keep the mathematical explanations light and focus on the *process*. You should include the insights collected from data analysis steps in order to handle the data and prepare well for model development.

- **Data Splitting:** State how you divided your data for training and testing.

Example: The data was split into 80% Training (759 records) and 20% Testing (190 records) to ensure the model generalizes well.

- **Model Selection & Training:** List the algorithms used and briefly show their default performance (e.g., Actual vs. Predicted plots).

Example: Three models were built to predict Productivity_Score: Linear Regression , Random Forest Regressor , and XGBoost...

V. Results and Evaluation (Secondary Focus)

Compare the models to determine which one learned the patterns in your data best.

- **Performance Metrics:** Compare the models using standard regression (RMSE, MAE,) or classification (Accuracy, F1-Score) metrics in a table or bar chart.
- **Comparison of models' performance.**
- **Feature Importance:** Which variables actually drove the model's decisions?

VI. Conclusion & Business Implications

Summarize the findings and translate them into actionable, real-world advice.

- **Key Findings:** What is the one major takeaway from your data?
- **Implications:** How can a business or user apply this knowledge?
- **Limitations & Future Work:** Acknowledge what your data couldn't tell you.

VII. References